Step 1 — A Script That Sends One Message and Prints the Reply — Solution

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Story — Step 1: before any chat, memory or agents, prove the three foundations work together: settings load from `.env`, logging works, and one call to the model comes back. Each piece goes in its final folder from day one, so later steps only *add* files. **If not:** a wrong key or a broken import would first show up in Step 3, mixed in with ticket logic, and you'd debug three things at once.

After this step the project already runs: `python main.py` sends one message and prints the reply.

The Doc01 files

These three files are Doc01's Build Task code (Intermediate, Approach 1), **unchanged** — only the file locations and the `from src...` import line differ.

Story — `src/exceptions.py`: `load_config()` needs a way to say "a required setting is missing" that's more specific than a bare `Exception`. **If not:** `main.py` would have to catch every `Exception`, which also hides real bugs.

```
# src/exceptions.py
# The project's named errors. MissingConfigError: a required setting is
# missing from .env. (Doc01's Build Task file.)
class MissingConfigError(Exception):
    """Raised when a required setting is missing from the environment."""
    pass
```

Story — `src/config.py`: the one place the project reads its settings from, failing loudly at startup. **If not:** every file would read `os.environ` itself, and a missing key would surface as a confusing `AuthenticationError` later.

```
# src/config.py
# Loads the settings from .env into a checked Config object. load_config()
# raises MissingConfigError when OPENAI_API_KEY is missing.
# (Doc01's Build Task file — only the import line changed.)
import os
from dataclasses import dataclass
```

```
from dotenv import load_dotenv
from src.exceptions import MissingConfigError

@dataclass
class Config:
    openai_api_key: str
    log_level: str

def require_env(key: str) -> str:
    value = os.getenv(key)
    if value is None or value == "":
        raise MissingConfigError(
            f"Required environment variable is missing: {key}"
        )
    return value

def load_config() -> Config:
    load_dotenv()
    api_key = require_env("OPENAI_API_KEY")
    log_level = os.getenv("LOG_LEVEL", "INFO")
    return Config(openai_api_key=api_key, log_level=log_level)
```

`load_dotenv()` looks for `.env` starting from this file's folder and walking up, so it finds the `.env` in the repo root.

Story — `src/utils/logger.py`: every file logs the same way, from one setup function. **If not:** each file would configure its own handler, and some lines would print twice.

```
# src/utils/logger.py
# get_logger(name): one logging setup for every file — a timestamp, the
# file name and the level on every line. (Doc01's Build Task file.)
import logging
import os

def get_logger(name: str) -> logging.Logger:
    logger = logging.getLogger(name)

    if not logger.handlers:
        level_name = os.getenv("LOG_LEVEL", "INFO")
        level = getattr(logging, level_name.upper(), logging.INFO)

        handler = logging.StreamHandler()
        handler.setLevel(level)
        log_format = "%(asctime)s %(name)s %(levelname)s %(message)s"
        formatter = logging.Formatter(log_format)
        handler.setFormatter(formatter)
```

```
        logger.addHandler(handler)
        logger.setLevel(level)

    return logger
```

Also create the empty `src/models/__init__.py` and `src/utils/__init__.py`, so both folders are importable packages.

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Solution — typed, logged, and the key checked at startup

Story — `models/llm.py`: the only file in the project that talks to the OpenAI SDK. It adds a `MODEL` constant and a startup key check. The key check sends a 1-token request, which costs almost nothing and tests the exact endpoint every agent will use. **If not:** a wrong key would first appear as a crash in the middle of the user's first real message.

```
# src/models/llm.py
# The only file that talks to the OpenAI SDK: create_client(), the startup
# key check, and send_message().
import openai
from openai import OpenAI

from src.config import Config

MODEL = "gpt-4o-mini"

def create_client(config: Config) -> OpenAI:
    return OpenAI(api_key=config.openai_api_key)

def check_api_key_at_startup(client: OpenAI) -> bool:
    try:
        client.chat.completions.create(
            model=MODEL,
            messages=[{"role": "user", "content": "ping"}],
            max_tokens=1,
        )
    except openai.AuthenticationError:
        return False
    return True

def send_message(client: OpenAI, history: list[dict], user_input: str) -> str:
    history.append({"role": "user", "content": user_input})
    response = client.chat.completions.create(model=MODEL, messages=history)
    reply = response.choices[0].message.content
    history.append({"role": "assistant", "content": reply})
    return reply
```

Story — `main.py`: the entry point. Settings first, then the logger, then the key check, then the one real call. Each failure gets one clear line and a clean exit. **If not:** a missing or wrong key would print

a long traceback that means nothing to a user.

```
# main.py
# The file you run: python main.py
# Loads the settings, checks the key, sends one message, prints the reply.
import sys

from src.config import load_config
from src.exceptions import MissingConfigError
from src.models.llm import (
    check_api_key_at_startup,
    create_client,
    send_message,
)
from src.utils.logger import get_logger

def main() -> None:
    try:
        config = load_config()
    except MissingConfigError as e:
        print(f"Setup problem: {e}")
        sys.exit(1)

    logger = get_logger(__name__)
    client = create_client(config)
    if not check_api_key_at_startup(client):
        logger.error("Authentication failed — check OPENAI_API_KEY.")
        print("Login failed — check OPENAI_API_KEY in your .env file.")
        sys.exit(1)

    logger.info("Sending a test message to the model")
    history: list[dict] = []
    reply = send_message(client, history, "What's 2+2?")
    print(reply)

if __name__ == "__main__":
    main()
```

Run it (from the repo root): `python main.py`.

Expected output (your timestamp and the model's wording will differ):

```
2026-09-25 10:00:00,123 __main__ INFO Sending a test message to the model
2 + 2 equals 4.
```

The logger names itself `__main__` here, because the file you run is always named `__main__`.

Check the failure paths:

- Remove `OPENAI_API_KEY` from `.env` → `Setup problem: Required environment variable is missing: OPENAI_API_KEY`
- Set it to `sk-fake-key-123` → an `ERROR` log line, then `Login failed — check OPENAI_API_KEY in your .env file.`

Both exit with code 1 and no traceback.

The Step 1 tests

Story — `tests/unit/test_config.py`: Doc01's own `test_config.py`, with only the imports changed — it proves a missing key stops the app with a clear error, with no API call. **If not:** the first sign of a broken `.env` would be a confusing OpenAI error.

```
# tests/unit/test_config.py
# Checks load_config() with no API call — Doc01's test_config.py, with
# the imports changed. It rewrites .env as it goes, so it moves your real
# .env aside first and puts it back at the end.
# Run from the repo root: python -m tests.unit.test_config
#   case 1 — no .env file: MissingConfigError
#   case 2 — .env without OPENAI_API_KEY: MissingConfigError
#   case 3 — a complete .env: the Config loads
import os

from src.config import load_config
from src.exceptions import MissingConfigError

ENV_PATH = ".env"
BACKUP_PATH = ".env.backup"


def set_env_file(contents: str | None) -> None:
    os.environ.pop("OPENAI_API_KEY", None)
    os.environ.pop("LOG_LEVEL", None)
    if contents is None:
        if os.path.exists(ENV_PATH):
            os.remove(ENV_PATH)
    else:
        with open(ENV_PATH, "w") as env_file:
            env_file.write(contents)


def main() -> None:
    if os.path.exists(ENV_PATH):
```

```
        os.rename(ENV_PATH, BACKUP_PATH)

    set_env_file(None)
    try:
        load_config()
    except MissingConfigError as e:
        print(f"case 1 (.env missing)  -> MissingConfigError: {e}")

    set_env_file("LOG_LEVEL=INFO\n")
    try:
        load_config()
    except MissingConfigError as e:
        print(f"case 2 (key missing)   -> MissingConfigError: {e}")

    set_env_file("OPENAI_API_KEY=sk-test-123\n")
    config = load_config()
    print(f"case 3 (valid .env)    -> loaded, log level {config.log_level}")

    if os.path.exists(ENV_PATH):
        os.remove(ENV_PATH)
    if os.path.exists(BACKUP_PATH):
        os.rename(BACKUP_PATH, ENV_PATH)


if __name__ == "__main__":
    main()
```

Story — `tests/integration/test_live_agents.py` : the checks that need the real model. It starts small — one real reply, and the startup key check — and every later step adds its own cases to this same file. It's the same style as Doc04's `test_chat_client.py` . **If not:** "it works" would only mean "it worked the one time I tried it by hand".

```
# tests/integration/test_live_agents.py
# The project's checks against the real OpenAI API — each step adds
# cases. Needs a valid OPENAI_API_KEY in .env.
# Run from the repo root: python -m tests.integration.test_live_agents
#   case 1 — one message in, a real reply out
#   case 2 — a wrong key is caught at startup, not mid-chat
from openai import OpenAI

from src.config import load_config
from src.models.llm import (
    check_api_key_at_startup,
    create_client,
    send_message,
)
```

```
def test_first_reply(client: OpenAI) -> None:
    reply = send_message(client, [], "What's 2+2?")
    print(f"case 1 (first reply)   -> {reply}")


def test_bad_key() -> None:
    bad_client = OpenAI(api_key="sk-fake-key-123")
    key_ok = check_api_key_at_startup(bad_client)
    print(f"case 2 (bad key ok?)   -> {key_ok}")


if __name__ == "__main__":
    config = load_config()
    client = create_client(config)
    test_first_reply(client)
    test_bad_key()
```

Also create the empty folders `tests/unit/` and `tests/integration/`. Tests run as modules, so no `__init__.py` is needed there.

Run the tests (from the repo root):

```
python -m tests.unit.test_config
python -m tests.integration.test_live_agents
```

Expected output:

```
case 1 (.env missing)  -> MissingConfigError: Required environment variable
is missing: OPENAI_API_KEY
case 2 (key missing)   -> MissingConfigError: Required environment variable
is missing: OPENAI_API_KEY
case 3 (valid .env)    -> loaded, log level INFO
case 1 (first reply)   -> 2 + 2 equals 4.
case 2 (bad key ok?)   -> False
```

(Cases 1 and 2 of the first file are shown wrapped onto 2 lines to fit the page — really one line each.)